Activity: ZIP models

The data in this class activity comes from a study on residents of Framingham, MA, which was conducted to research variables related to heart health. We will work with a subset of the data, containing

- `cigsPerDay`: The number of cigarettes smoked per day during the study period.
- `education`: 1 = High School, 2 = Some College, 3 = College Degree, 4 = Advanced Degree.
- `male`: 1 = Male, 0 = Female.
- `age`: The age of the individual in years.
- `diabetes`: 1 if the individual has diabetes, 0 otherwise.

While the data were originally collected to study heart health, in this activity we will try to model the number of cigarettes smoked.

The model

To account for excess zeros in the data, we fit a *zero-inflated Poisson* (ZIP) model. Here is the probability function for our model:

$$P(Y_i = y|\mathbf{x}_i) = \begin{cases} e^{-\lambda_i}(1-p_i) + p_i & y = 0 \\ \dfrac{e^{-\lambda_i}\lambda_i^y}{y!}(1-p_i) & y > 0 \end{cases}$$

where

$$\log\left(\frac{p_i}{1-p_i}\right) = \mathbf{x}_i^T\gamma$$
$$\log(\lambda_i) = \mathbf{x}_i^T\beta$$

Model output

```
m1 <- zeroinfl(cigsPerDay ~ male + age + education + diabetes,
               dist = "poisson", data = heart_data)

summary(m1)
```

```
Call:
zeroinfl(formula = cigsPerDay ~ male + age + education + diabetes, data = heart_data,
    dist = "poisson")

Pearson residuals:
    Min      1Q  Median      3Q     Max
-1.8545 -0.8650 -0.5560  0.6747  6.0641

Count model coefficients (poisson with log link):
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  2.9585841  0.0350454  84.422   <2e-16 ***
male         0.4621809  0.0109838  42.078   <2e-16 ***
age         -0.0067425  0.0006782  -9.941   <2e-16 ***
education2   0.0131422  0.0125969   1.043    0.297
education3   0.0128553  0.0160200   0.802    0.422
education4  -0.0187548  0.0171957  -1.091    0.275
diabetes    -0.0166835  0.0393369  -0.424    0.671

Zero-inflation model coefficients (binomial with logit link):
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.139429   0.215937  -9.908   <2e-16 ***
male        -0.846393   0.067171 -12.601   <2e-16 ***
age          0.051276   0.004049  12.664   <2e-16 ***
education2  -0.130213   0.080576  -1.616    0.106
education3   0.055617   0.095845   0.580    0.562
education4   0.141369   0.110157   1.283    0.199
diabetes     0.337119   0.215137   1.567    0.117
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Number of iterations in BFGS optimization: 21
Log-likelihood: -1.297e+04 on 14 Df
```

Questions

1. Why might a ZIP model be appropriate for this data? What would the latent variable represent?

2. Interpret the coefficient 0.462 in the fitted model.